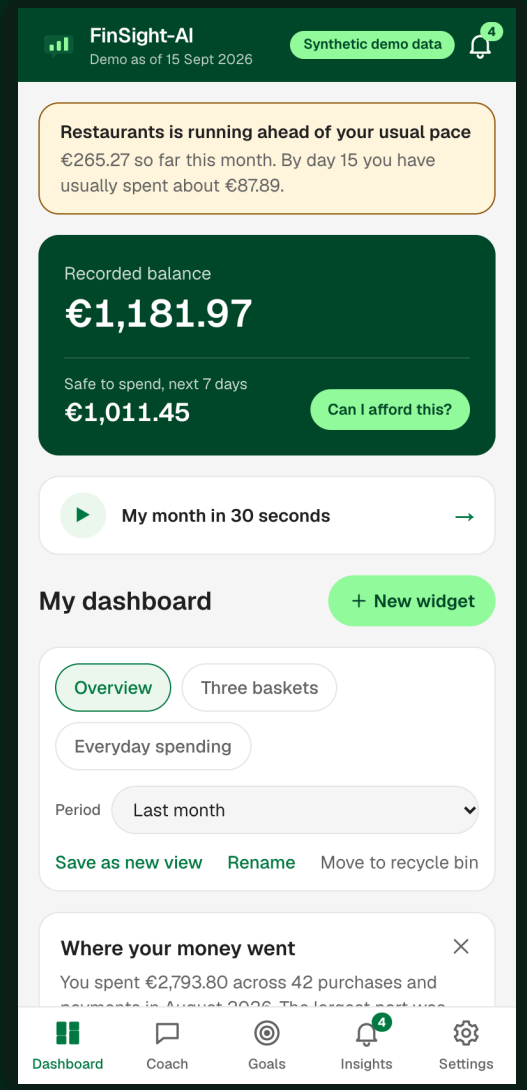
How FinSight-AI works out its answers

The logic, the formulas and the statistics behind the coach



The AI chooses, the code calculates, a check verifies

1 The AI chooses → **2 The code calculates** → **3 A check verifies**

It reads the question and picks one of 16 calculations. It sees totals, never transactions.

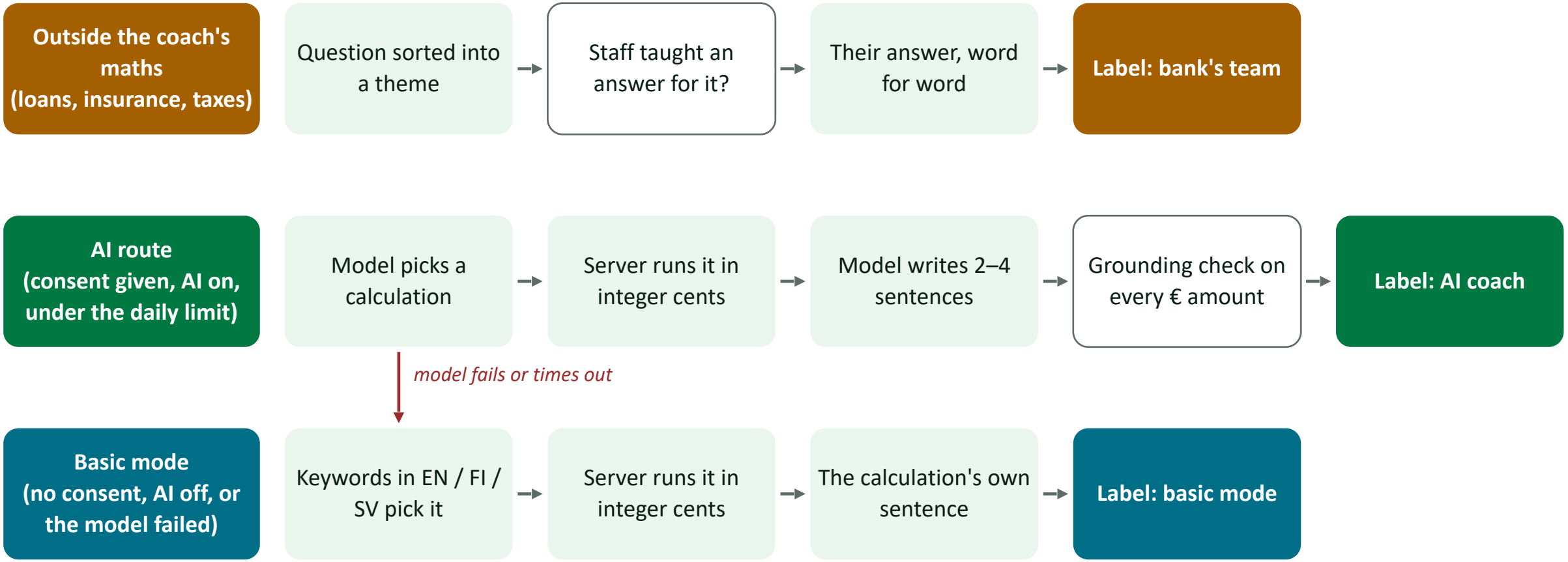
Deterministic code runs on our server in integer cents. Same input, same answer, every time.

Every euro amount in the AI's wording must be one the calculation produced, or the wording is dropped.

$$\text{answer shown} \Leftrightarrow \{\,€\text{ amounts in the wording}\,\} \subseteq \{\,€\text{ amounts the calculation produced}\,\}$$

With AI switched off, declined or failing, keywords pick the calculation instead. Steps 2 and the charts are identical.

From a question to an answer: three routes



The customer always gets an answer, and every answer says which route produced it.

The grounding check keeps the AI honest

1

Find every amount in the wording

€1,011.45 · 1 011,45 € · 300 euros

2

Parse each to integer cents

1 011,45 € → 101145

3

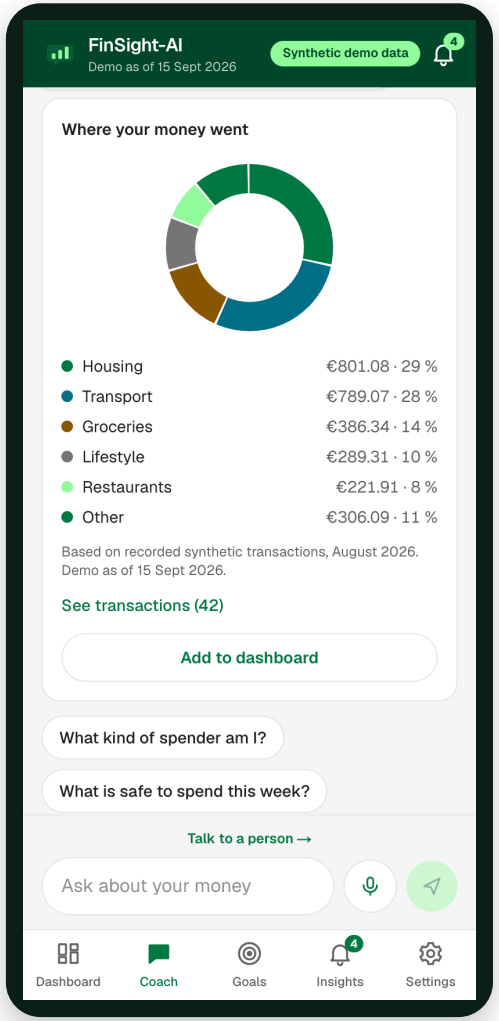
Look it up in the calculated set

exact, or the same figure rounded to whole euros

All supported
→ show the AI's wording

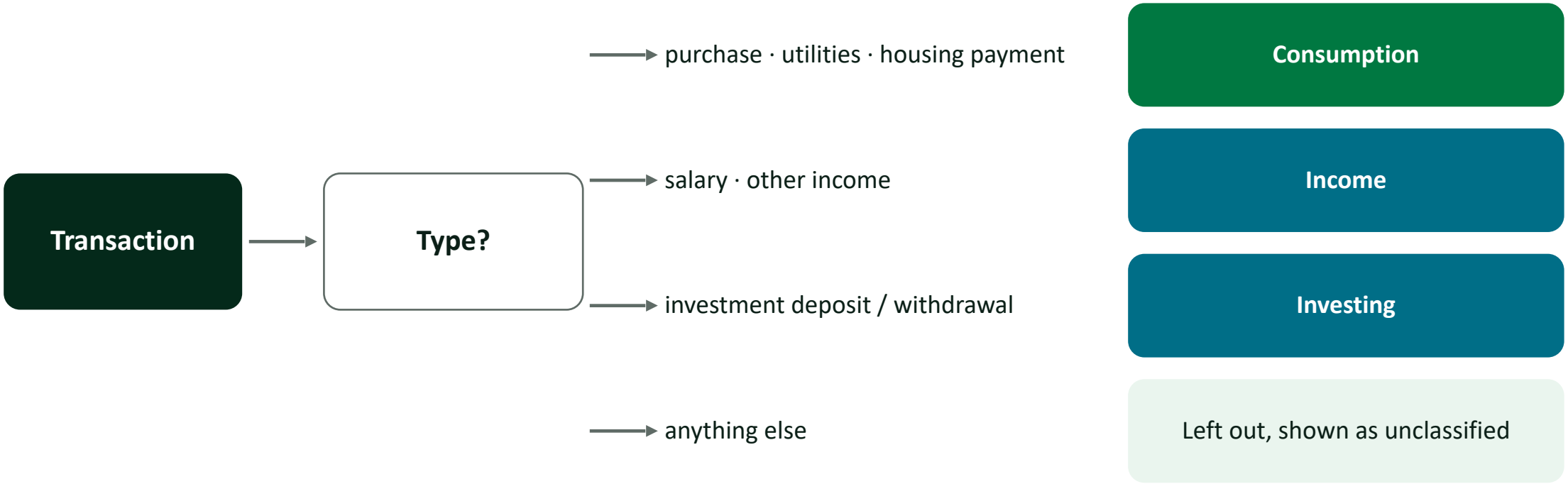
Any unsupported
→ use the calculation's sentence

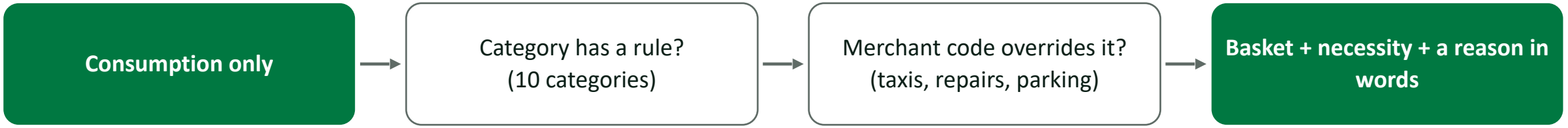
$$\text{grounded} \Leftrightarrow \forall\, a \in A(\text{text}):\ a \in S \ \vee\ a = \text{round€}(s),\ s \in S$$



Every figure opens its transactions

How a transaction is categorised





The three baskets, and how necessary each purchase is

Category	Basket	Necessity	Merchant-code override
Housing, Utilities	Regular bills	Essential	—
Health	Irregular essentials	Essential	—
Home & Tech	Irregular essentials	Unknown	Home supplies, hardware → essential
Groceries	Everyday spending	Essential	—
Transportation	Everyday spending	Essential	Taxis → free choice · parking → unknown
Restaurants, Entertainment	Everyday spending	Free choice	—
Lifestyle, Travel	Everyday spending	Free choice	—

Essentials are never suggested as savings

“Unknown” is said out loud, not guessed

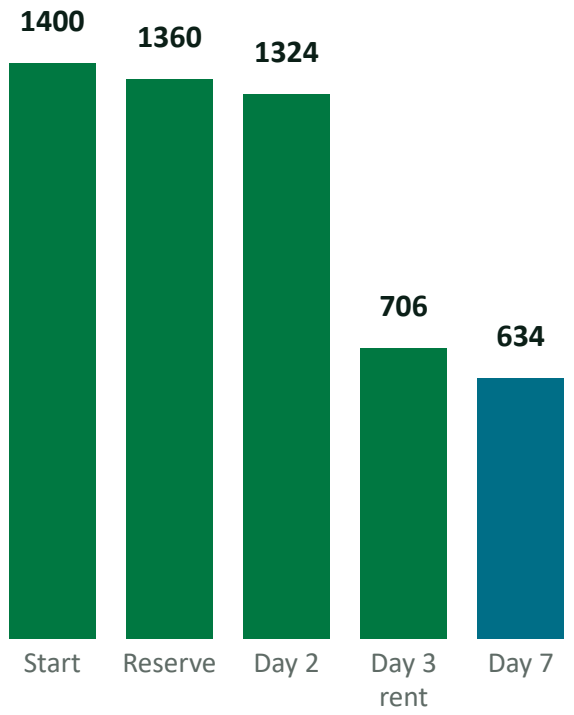
The customer's own correction wins

Safe to spend is the lowest point of a day-by-day walk

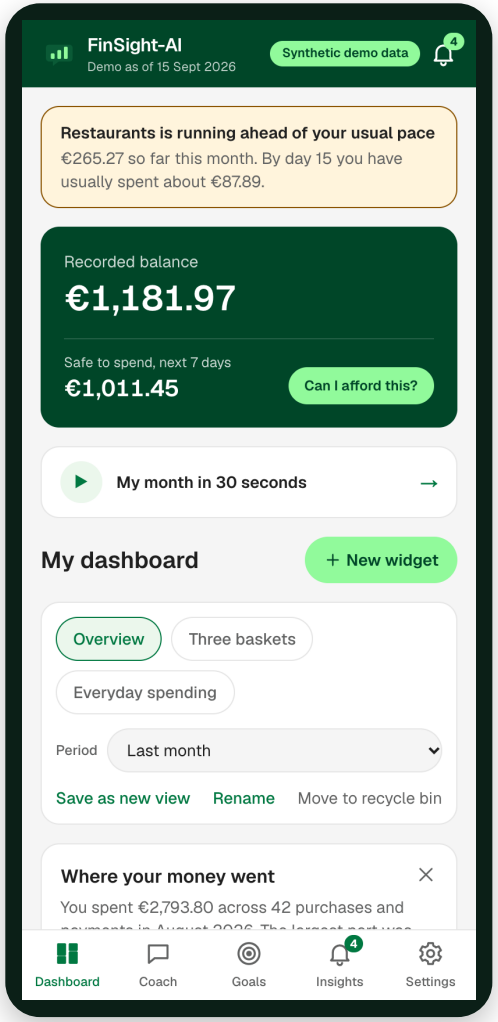
$$\text{safe} = \max(\,0\,,\ \min \text{ over days d of } H(d)\,)$$
$$H(d) = B - P - R - X - \Sigma\,(e + s + \text{bills}_i),\ i = 1 \ldots d$$

B	recorded balance
P	protected: money set aside for goals
R	reserve: mean monthly irregular essentials × days ÷ 30
X	a planned purchase (“can I afford…?”)
e	everyday essentials per day (3-month average)
s	bills spread over the month, per day
bills_i	bills that fall due on day i

Worked example: headroom, €



Lowest point = safe to spend: €634



Finding regular bills, paydays and subscriptions

Regular bills

- Regular-basket category, last 6 full months
- Seen in ≥ 3 of the months
- Amount = median monthly total
- One payment a month → day = median day
- Several a month → spread: monthly ÷ 30 a day

Payday

- Salary transactions, last 6 full months
- Days grouped: 1st, end of month, or nearest 5
- A group seen ≥ 3 times is a payday
- Amount = median salary
- Not counted in safe to spend until it arrives

Subscriptions

- Same merchant, last 6 full months
- Exactly one payment in a month
- Seen in ≥ 3 months
- (max − min) ÷ median ≤ 10 %
- Yearly cost = 12 × median amount

next expected date = the typical day, in the first month where that day is still ahead

What-if, goals and safe to save

What if I cut restaurants by 20 %?

saving = baseline × p
per year = saving × 12

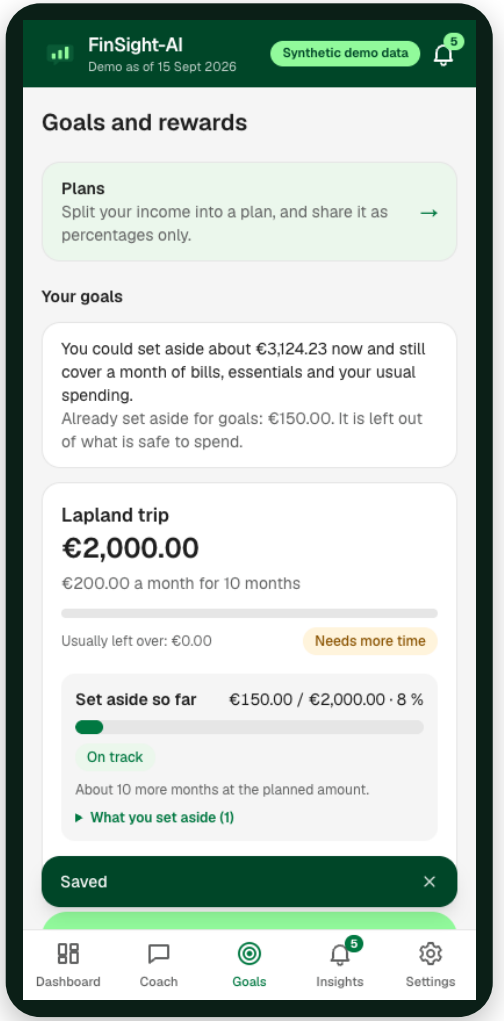
Can I reach this goal?

monthly = ⌈ target ÷ months ⌉
room = mean income − mean spending (6 months)
gap = max(0 , monthly − room)

How much can I save?

safe to save = max(0 , H_{30} − usual free-choice month)
progress = set aside ÷ target

Room covers it → fits → **Trims of 10–30 % close it → a stretch** → **Still short → say so**



Peer benchmarks: the middle half of your age group

5,100 customers · 3.6 M transactions

↓

Per customer: average month over 12 full months, per metric

↓

Per age band: 25th, 50th, 75th percentile (nearest rank)

↓

Fewer than 30 in a group → dropped

↓

Stored: percentiles only. No row is a person

Where a customer sits in their group



$$\text{position} = \begin{cases} \text{lower} & \text{if } x < P25 \\ \text{typical} & \text{if } P25 \le x \le P75 \\ \text{higher} & \text{if } x > P75 \end{cases}$$

Used in: the “people my age” answer, one insight (above P75), one recommendation (move halfway to P50), one admin chart.

Statistics Finland gives national context, and decides nothing

Source	Statistics Finland, table 14q4
Year	2022
Measure	€ per month per consumption unit
Licence	CC BY 4.0

Shown as context only.
Never used to rank, alert or recommend.

Our category	National (COICOP) division
Groceries	01 Food and non-alcoholic drinks
Health	06 Health
Transportation	07 Transport
Entertainment	09 Recreation and culture
Restaurants	11 Restaurants and accommodation

One account is not a whole household and the categories differ slightly, so the app says: read this as context, not a ranking.

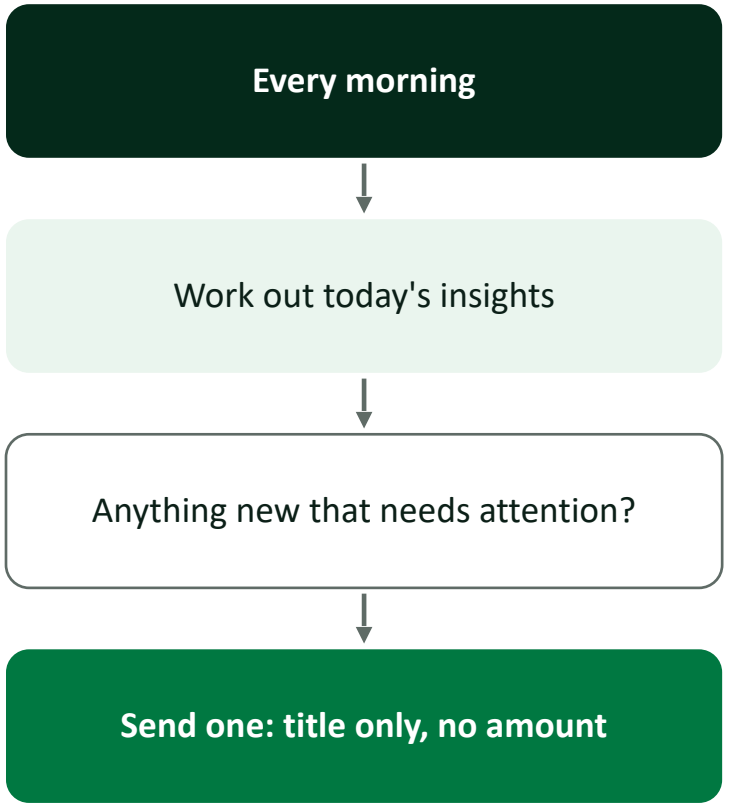
Recommendations: six rules, ranked by monthly value

#	Fires when	Suggestion	Monthly value
1	A free-choice category is above the age group's P75	Move halfway to the middle	$(\text{own} - \text{P50}) \div 2$
2	Largest free-choice category is over €50 a month	A modest trim	15 % of it
3	8 or more free-choice purchases under €15 last month	Notice the small ones	25 % of their total
4	Essential irregular costs in the last 12 months	Reserve of the 12-month total ÷ 12	protection
5	Balance below one month of regular bills	Build a one-month buffer	protection
6	Over €100 left each month and nothing invested	Put the surplus to work	protection

Rewards use the same data, and compare a customer only with their own earlier months, never with other people.

Insights and reminders: what triggers each one

Insight	Fires when
Possible shortfall	The 7-day walk goes below zero
Thin buffer	Balance < one month of regular bills
Habit going the wrong way	From day 7: free-choice category ≥ €30 and ≥ 25 % ahead of usual
Bill due	A regular bill is expected within 10 days
Category spike or drop	Last month ± 25 % and ± €30 vs its 3-month mean
Small purchases add up	8+ free-choice purchases under €15
Above the age group	A free-choice category above P75
Goal reminder	Once a month, per goal



A plan is five percentages, so it can be shared without amounts

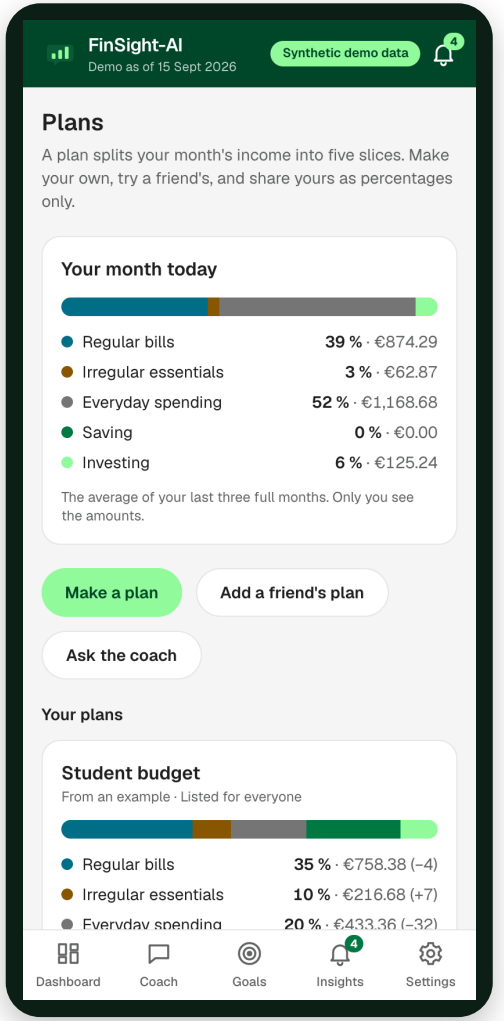
$$p_i = \lfloor\, 100 \cdot a_i \div \Sigma a \,\rfloor\,, \text{ then } +1 \text{ to the largest remainders until } \Sigma\, p_i = 100$$

Worked example: income €2,500 a month

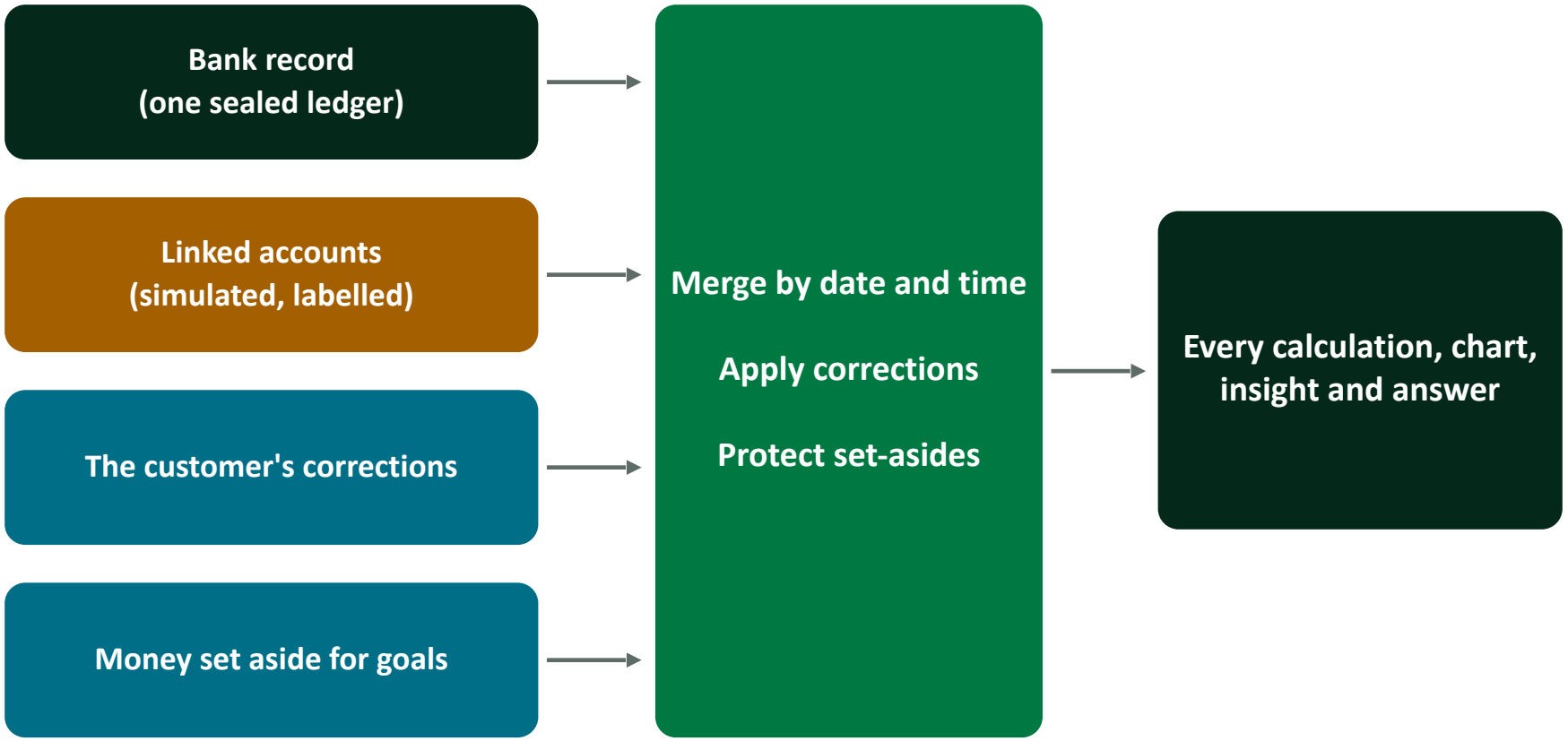
Slice	€	Exact %	Floor	Final
Regular bills	900	36.0	36	36
Everyday spending	655	26.2	26	26
Irregular essentials	70	2.8	2	3
Saving	875	35.0	35	35

Others see: the name and five percentages

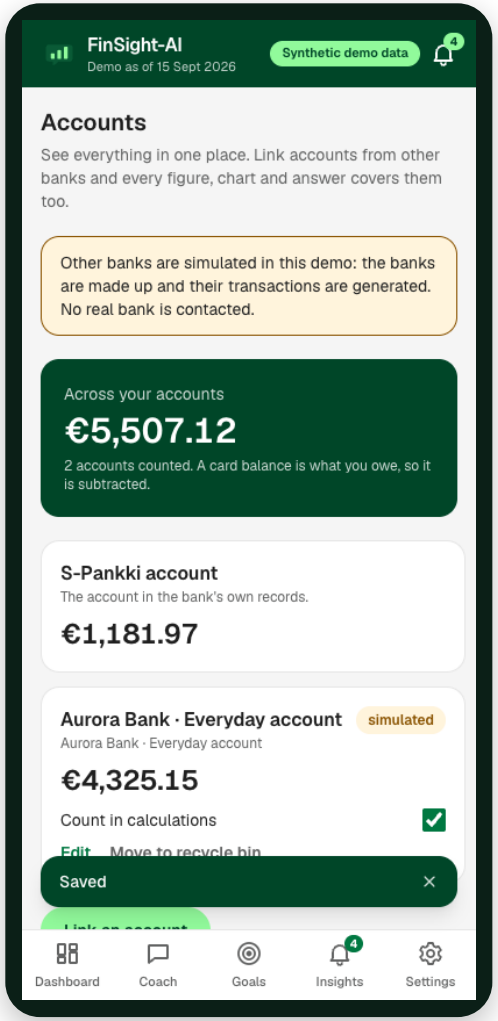
Never: income, any amount, balance, name, age, customer number



One picture of the customer, built before any calculation runs



combined balance = Σ latest balance of each counted account



The mortgage guide teaches from 200 applications

25 days
typical length (13–45)

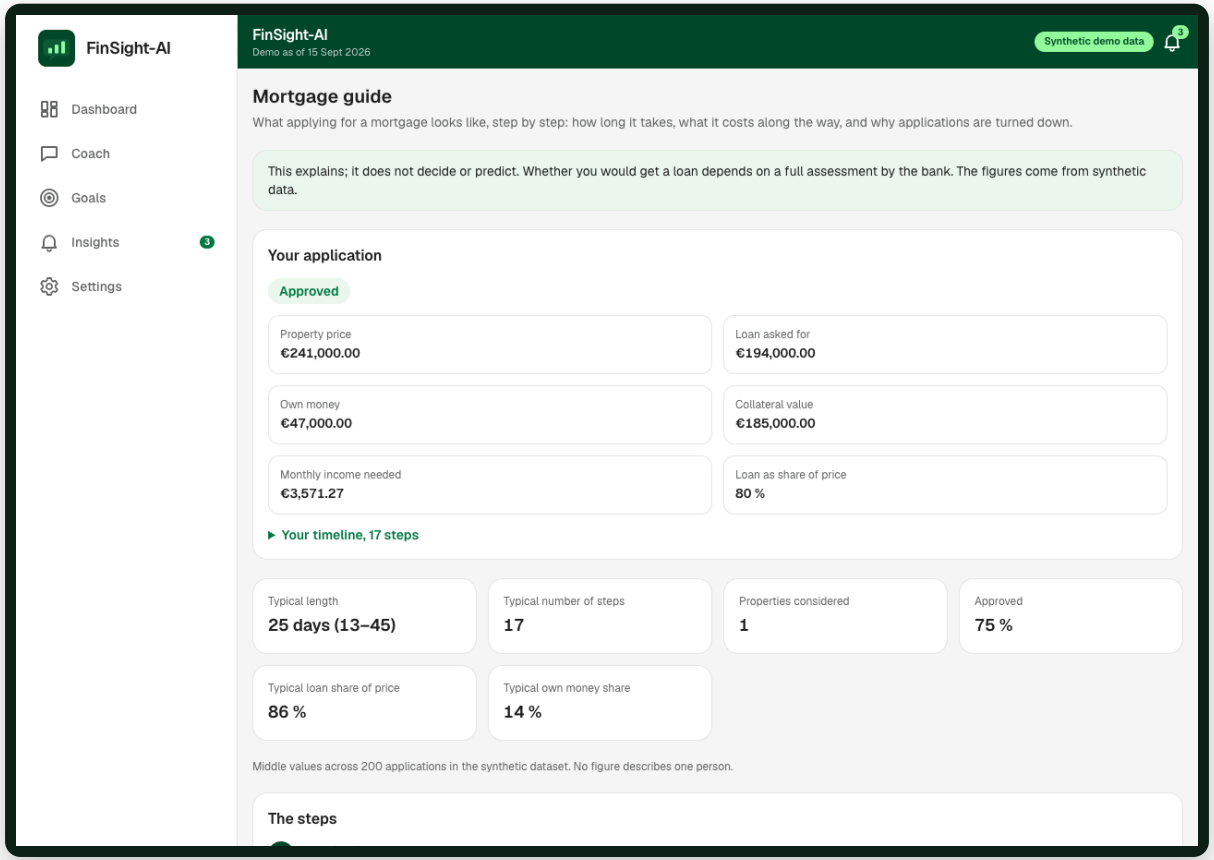
17
typical number of steps

75 %
approved

86 %
typical loan share of price

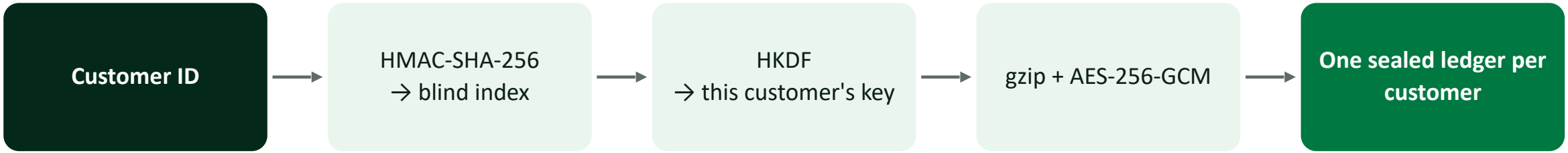
each figure = median across applications
step day = median day of first occurrence

Explains. Never predicts approval.



Why not approved, among the 50: collateral 68 %, income 16 %, own money 12 %, existing debt 4 %.

Nothing a customer owns is stored unencrypted



ref = HMAC(index key, customer ID) key = HKDF(master key, ref)
sealed = AES-256-GCM(key, gzip(ledger), context)

The AI sees	The AI never sees	Staff see
The question, recent turns and totals. Only with consent.	Transactions, customer ID, age, postal code.	Counts by topic and day. Never who asked or what they typed.

What is behind the answers, in numbers

5,100
customers loaded

3,626,755
transactions, sealed per customer

16
calculations the coach can run

15
chart types to draw them

3
languages, text and voice

100
automated tests

Advice you can check: every answer opens the transactions it was calculated from.